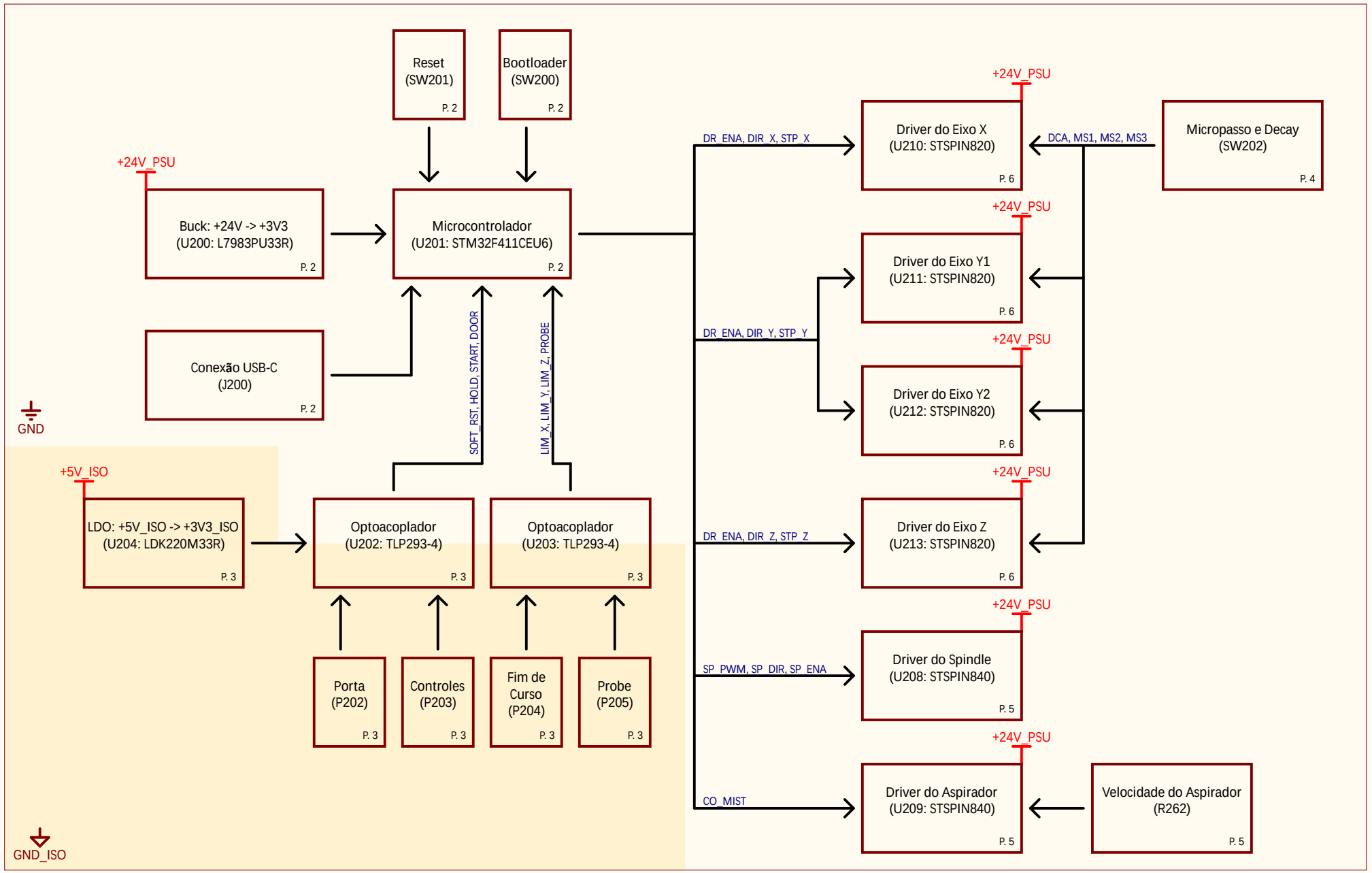
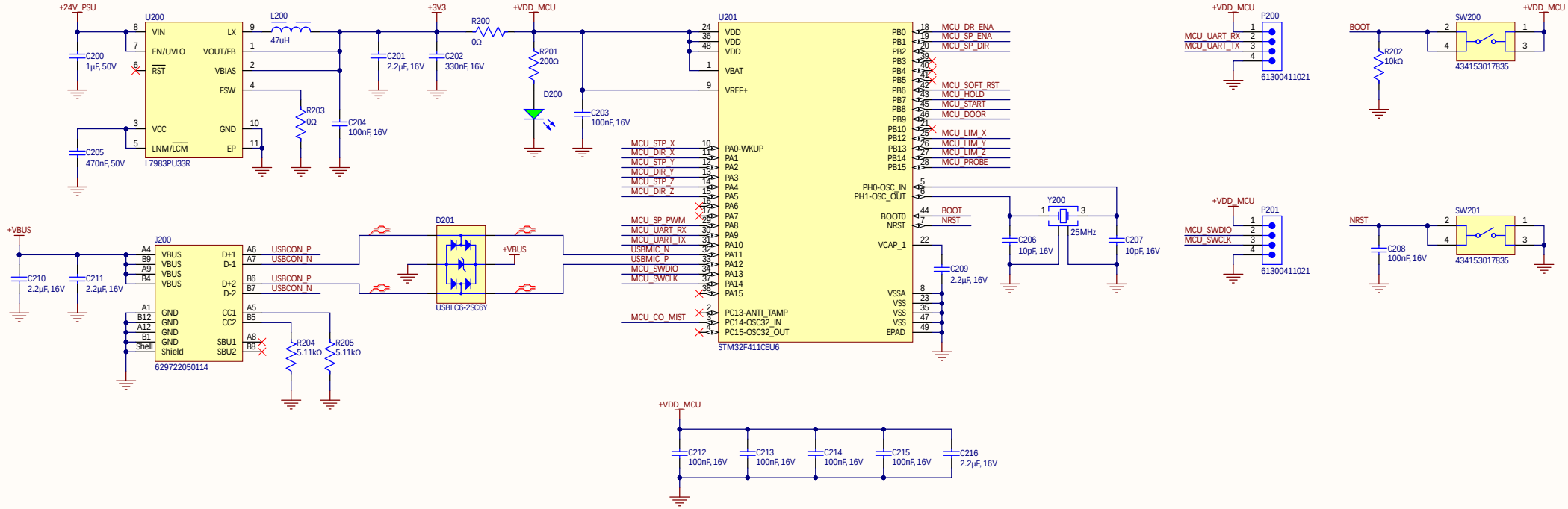

Reset
(SW201)
P. 2
Bootloader
(SW200)
P. 2
+24V_PSU
Buck: +24V -> +3V3
(U200: L7983PU33R)
P. 2
Microcontrolador
(U201: STM32F411CEU6)
P. 2
Conexão USB-C
(J200)
P. 2
GND
SOFT_RST, HOLD, START, DOOR
LIM_X, LIM_Y, LIM_Z, PROBE
+5V_ISO
LDO: +5V_ISO -> +3V3_ISO
(U204: LDK220M33R)
P. 3
Optoacoplador
(U202: TLP293-4)
P. 3
Optoacoplador
(U203: TLP293-4)
P. 3
Porta
(P202)
P. 3
Controles
(P203)
P. 3
Fim de
Curso
(P204)
P. 3
Probe
(P205)
P. 3
GND_ISO
DR_ENA, DIR_X, STP_X
DR_ENA, DIR_Y, STP_Y
DR_ENA, DIR_Z, STP_Z
SP_PWM, SP_DIR, SP_ENA
CO_MIST
+24V_PSU
Driver do Eixo X
(U210: STSPIN820)
P. 6
DCA, MS1, MS2, MS3
Micropasso e Decay
(SW202)
P. 4
Driver do Eixo Y1
(U211: STSPIN820)
P. 6
Driver do Eixo Y2
(U212: STSPIN820)
P. 6
Driver do Eixo Z
(U213: STSPIN820)
P. 6
Driver do Spindle
(U208: STSPIN840)
P. 5
Driver do Aspirador
(U209: STSPIN840)
P. 5
Velocidade do Aspirador
(R262)
P. 5

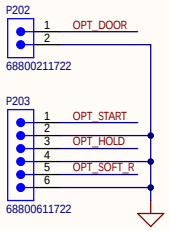
MICROCONTROLADOR

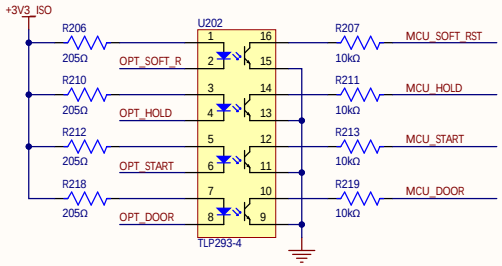


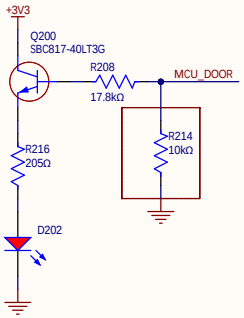
Title: Interface USB e Microcontrolador			
Size: Tabloid	Number: 2	Sheet of: 6	Revision: 0.1
Date: 02/11/2026	Drawn By: Sérgio Beltrão		
Project Name: DRIVER.PrjPcb			

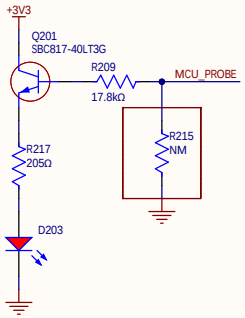
GSN

INTERFACES EXTERNAS



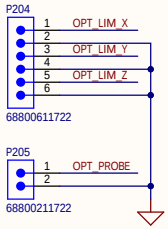


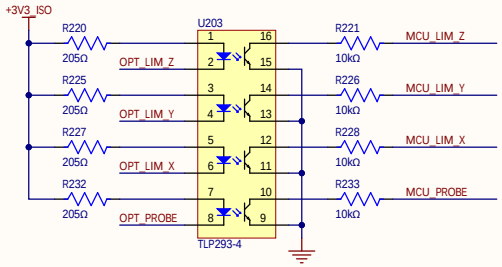


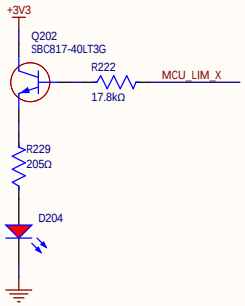


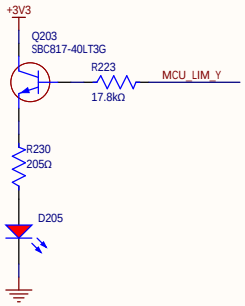
NOTA DE PROJETO

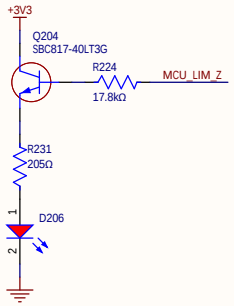
Caso as entradas Door e Probe não sejam utilizadas, instale resistores de 10 kΩ nas posições R214 e R215 para desabilitá-las fisicamente.

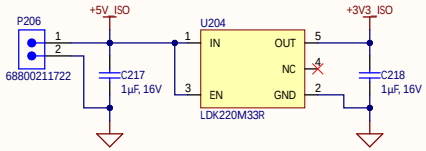








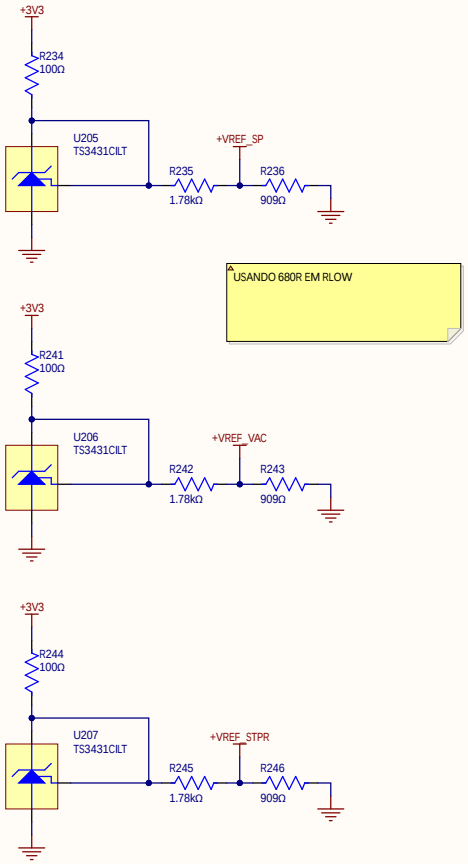


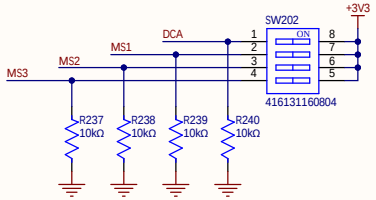


Title: Interfaces Externas			
Size: Tabloid	Number: 3	Sheet of: 6	Revision: 0.1
Date: 02/11/2026	Drawn By: Sérgio Beltrão		
Project Name: DRIVER.PrjPcb			

GSN

INTERFACE DOS DRIVERS





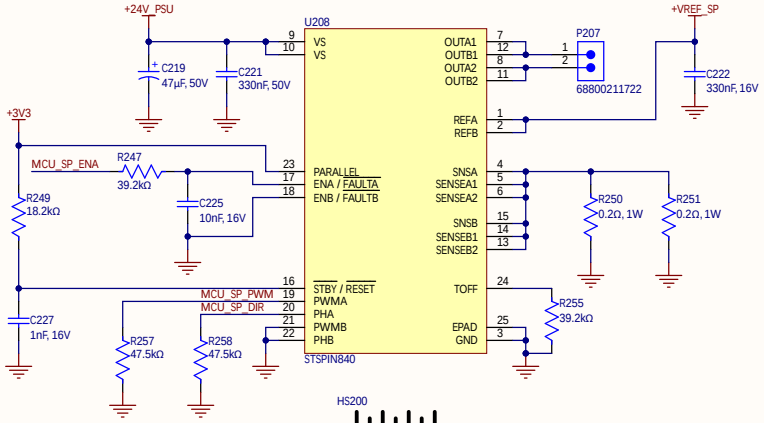
SELEÇÃO DO MODO DE DECAY	
SW1-1	MODO
DESLIGADO	MISTO
LIGADO	LENTO

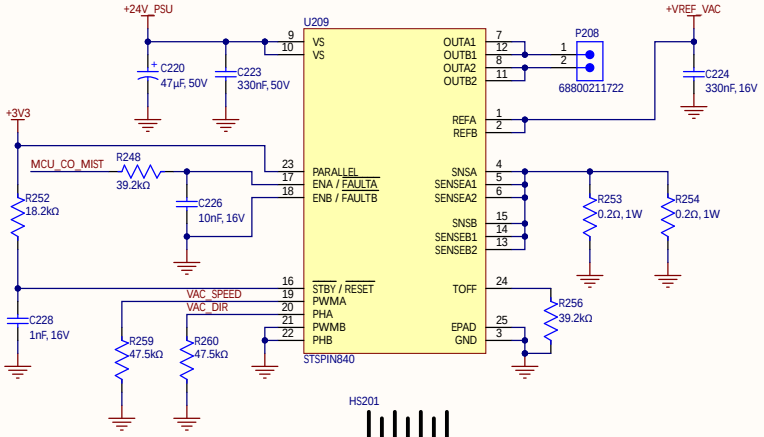
SELEÇÃO DO MODO DE PASSO			
SW1-2	SW1-3	SW1-4	PASSO
DESLIGADO	DESLIGADO	DESLIGADO	COMPLETO
LIGADO	DESLIGADO	DESLIGADO	1/2
DESLIGADO	LIGADO	DESLIGADO	1/4
LIGADO	LIGADO	DESLIGADO	1/8
DESLIGADO	DESLIGADO	LIGADO	1/16
LIGADO	DESLIGADO	LIGADO	1/32
DESLIGADO	LIGADO	LIGADO	1/128
LIGADO	LIGADO	LIGADO	1/256

Title: VREF e Seleção de Micropassos			
Size: Tabloid	Number: 4	Sheet of: 6	Revision: 0.1
Date: 02/11/2026	Drawn By: Sérgio Beltrão		
Project Name: DRIVER.PrjPcb			

GSN

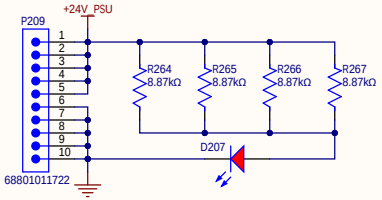
ACIONAMENTO DOS MOTORES DC





NOTA DE PROJETO

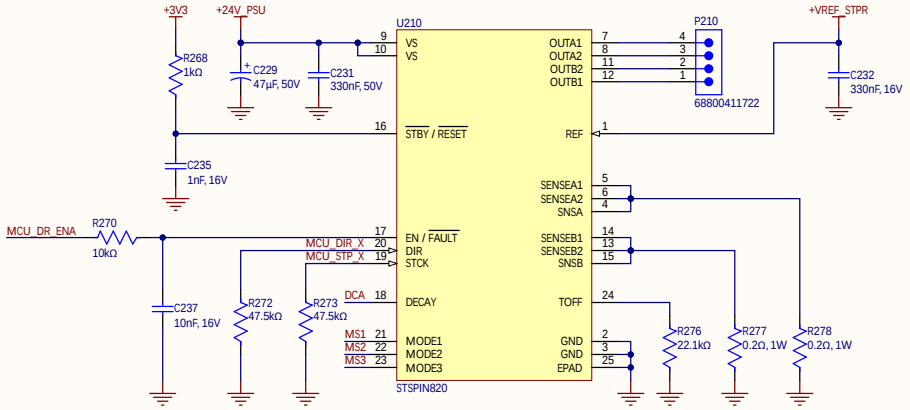
Monte R292 para inverter o sentido de rotação do motor do aspirador.

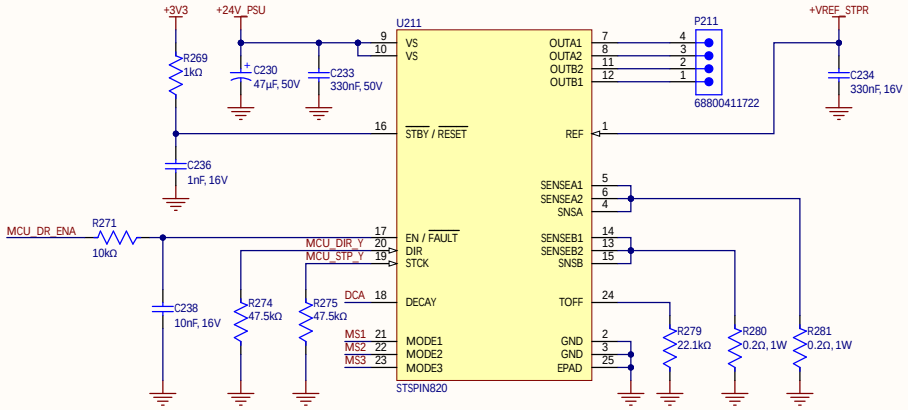


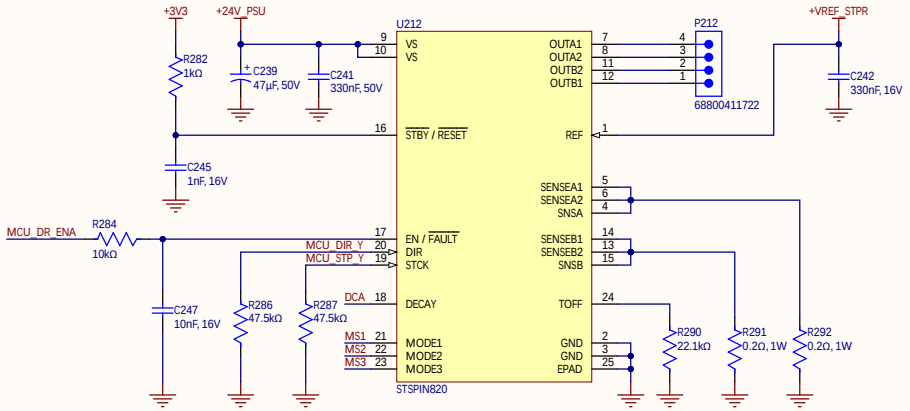
Title: Controle do Spindle e Aspirador			
Size: Tabloid	Number: 5	Sheet of: 6	Revision: 0.1
Date: 02/11/2026	Drawn By: Sérgio Beltrão		
Project Name: DRIVER.PrjPcb			

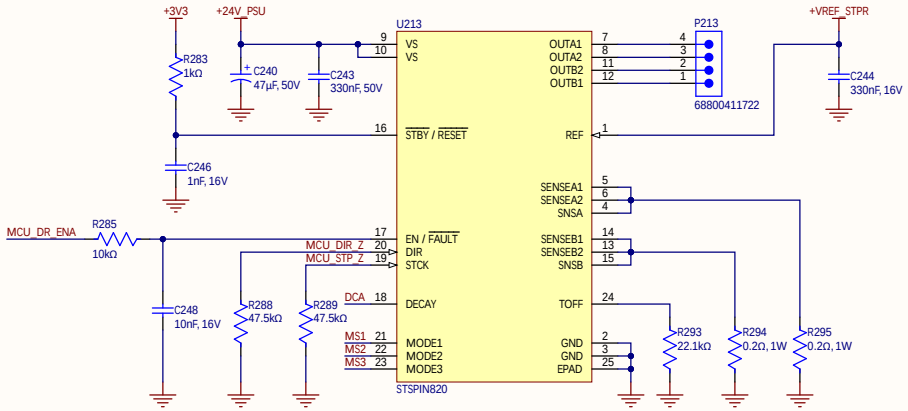
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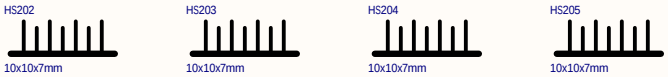
ACIONAMENTO DOS MOTORES DE PASSO











Title: Drivers de Motores de Passo			
Size: Tabloid	Number: 6	Sheet of: 6	Revision: 0.1
Date: 02/11/2026	Drawn By: Sérgio Beltrão		
Project Name: DRIVER.PrjPcb			

GSN